APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

Hull-White 1-Factor Rate Model

Model ID	APEX-MDL-0005
Version	1.0
Owner	Dr. Yuki Tanaka (Quant Researcher)
Validator	Dr. Samuel Achebe (Model Validation Officer)
Status	In Validation
Last Review	2026-03-01
Next Review	2027-03-01

Regulatory Framework: SR 11-7, FRTB delta-IR sensitivity

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1 Model Overview

1.1 Purpose

The Hull-White 1-Factor (HW1F) model is an arbitrage-free short-rate model used for pricing and risk-managing interest rate derivatives. It perfectly fits the initial yield curve by construction and provides closed-form solutions for zero-coupon bonds and European swaptions.

1.2 Business Use

- **IRS valuation:** Fixed-float leg decomposition using HW1F zero-coupon bond prices.
- **PFE simulation:** Monte Carlo IR path generation for XVA exposure profiles.
- **FRTB-SA IR delta:** Sensitivity bumps applied along HW1F term structure.
- **Option pricing:** Cap/floor and swaption valuations for rates desk.

2 Theoretical Basis

2.1 Short-Rate Dynamics

Under the risk-neutral measure $\mathbb{Q}$:

$$dr_t = \big[\theta(t) - a\, r_t\big]\, dt + \sigma\, dW_t$$

where a is the mean-reversion speed, σ is the short-rate volatility, and $\theta(t)$ is the time-dependent drift calibrated to match the initial term structure.

2.2 Analytical Zero-Coupon Bond

$$P(t,T) = A(t,T)\, \exp\big[-B(t,T)\, r_t\big]$$

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

$$\ln A(t,T) = \ln \frac{P_{\text{mkt}}(0,T)}{P_{\text{mkt}}(0,t)} - B(t,T)\, F(0,t) - \frac{\sigma^2}{4a}\big(1 - e^{-2at}\big) B(t,T)^2$$

where $F(0,t) = -\partial \ln P_{\text{mkt}}(0,t)/\partial t$ is the instantaneous forward rate.

2.3 IRS Valuation

$$V_{\text{fixed}} = N\, R \sum_i \tau_i\, P(0,T_i)$$

$$V_{\text{float}} = N\big(P(0,T_0) - P(0,T_n)\big)$$

$$V_{\text{IRS}} = V_{\text{float}} - V_{\text{fixed}} \quad \text{(payer)}$$

2.4 European Payer Swaption

$$V_{\text{swaption}} = \sum_i c_i \times \text{ZBP}(t,\, S,\, T_i,\, K_i)$$

where ZBP is the Hull-White zero-coupon bond put formula with effective strike adjustment.

3 Mathematical Specification

Parameter	Calibrated Value	Calibration Target
Mean reversion a	0.08	10Y IRS rate stability
Short-rate vol σ	0.012	ATM swaption vol
Calibration instruments	US2Y, US10Y live prices	OIS curve from market feed

4 Implementation

Code location: Referenced in `infrastructure/trading/greeks.py` for IRS DV01 computation; full HW1F path generation in PFE engine (`infrastructure/xva/adapter.py`).

IRS DV01 approximation:

$$\text{DV01}_{\text{IRS}} = \text{quantity} \times 0.0004 \quad \text{(USD per 1bp per \$1 notional)}$$

This is a flat DV01 approximation; HW1F gives the full term-structure sensitivity used for FRTB.

Calibration procedure: Daily calibration to US2Y and US10Y prices from `FeedHandler`. a calibrated to fit the ATM 2Y×5Y swaption vol target; σ fitted to replicate 10Y yield level.

5 Validation

Benchmark: IRS fair value compared against QuantLib HW1F; within ±0.01% of notional for par swaps.

Swaption validation: ATM swaption prices within ±2 bps implied vol of market references.

PFE profile check: 5Y USD IRS expected exposure profile: peak at ∼3Y (18% of notional), declines to zero at maturity. Consistent with industry benchmarks.

6 Model Limitations

1. **Single factor:** HW1F captures only parallel yield curve shifts. Twist and curvature moves not captured; error for 2Y vs 10Y movements can exceed 15% of DV01.
2. **Constant a and σ:** Time-homogeneous parameters mis-price long-dated swaptions and fail to reproduce the volatility hump at 2–3 year expiry.
3. **Gaussian rates:** HW1F allows negative rates; may mis-price floors at zero strike.
4. **OIS calibration only:** Calibrated to USD OIS; USD LIBOR/SOFR basis not modelled.

7 Open Findings

ID	Severity	Description	Status
HW1F-F1	**Major**	Single-factor limitation: twist and curvature yield curve moves not captured; material error for butterfly swaps and CMS products.	Open
HW1F-F2	**Minor**	Calibration uses only US2Y and US10Y; US5Y/US30Y excluded from calibration objective.	Open